

The Core Difference in your Design

RX600 32-Bit Microcontrollers



The RX architecture is future oriented and feature rich delivers superior performance in terms of core processing performance, code efficiency, and power consumption. CPU cores in the RX family (RXv1 and RXv2) marry the speed of a RISC architecture with the flexibility and code efficiency of a CISC architecture.

Features

Real Time Performance

- Up to 120MHz zero wait state flash
- Ultra fast 5 cycle interrupt response
- On chip FPU and DSP
- Up to 2 DMIPs/MHz
- Up to 4.25 Coremark/MHz

Broad Scalability

- 48 ~ 177 pins packages
- 32Kbytes ~ 4Mbytes Flash
- Core and peripheral compatibility

System Approach

- Enhanced system functionality
- Connectivity (Ethernet, Dual-USB, 3xCAN and more)
- Advanced timers and analog

Low Power, Low Voltage

- 500uA/MHz, with all the peripherals active
- 1.4uA RTC Deep Standby (RX631/63N)
- 1mW per DMIPS
- Extends battery life in portable applications

Quality and Safety

- Best-in-class ESD
- IEC60730 safety functions
- Electromagnetic susceptibility up to 500V immunity

Extensive software and tool support

- Industry-standard and 3rd Party Tools support
- Solution Kits (DD-LCDs, Wifi, Audio, Cloud Connectivity, Motor Control and more)

Applications

- Access Control
- IOT (Internet of Things - Cloud Connectivity)
- HMI (Human Machine Interface)
- Motor Control

Tools



RX for Connectivity and Motor Control

100MHz

RX63N

Ethernet CAN USB
LCD Direct Drive

RX631

CAN USB
LCD Direct Drive

RX63T

CAN LIN
Advanced Motor Control

120MHz

RX64M

Up to
4 MB Flash
512 KB SRAM
Dual Ethernet
Dual USB
3xCAN
LCD Direct Drive
Camera Interface

	Tools Product Name	Part Number
RX631 RX63N	Renesas Starter Kit* for RX631/RX63N-256K (Kit with Renesas CubeSuite+)	R0K50563NS010BE
	Renesas Starter Kit* for RX631/RX63N-256K (Kit with e2studio)	YR0K50563NS010BE
RX63T	Renesas Starter Kit for RX63T (64-pin)	R0K50563TS000BE
	Renesas Starter Kit* for RX63T(144-pin) (Kit with Renesas CubeSuite+)	R0K5563THS000BE
	Renesas Starter Kit* for RX63T(144-pin) (Kit with e2studio)	YR0K5563THS000BE
RX64M	Renesas Starter Kit* for RX64M (Supporting Trusted Memory) (Kit with Cube Suite+)	R0K50564MS000BE
	Renesas Starter Kit* for RX64M (Supporting Trusted Memory) (Kit with e2studio)	YR0K50564MS000BE

Renesas Electronics India Pvt. Ltd.



<http://in.renesas.com/rx600>

RENESAS